

## INF2006 CLOUD COMPUTING &amp; BIG DATA

# Team Project 1

## Secure, Scalable and Data-Driven Cloud Service

This project asks each team to design, build, test and justify a cloud-hosted digital service that is secure, scalable and informed by data. It is intentionally evidence-led: your submission must enable a reviewer to reproduce, inspect and assess what you built without relying on a live cloud account, a verbal explanation or an unstructured video.

### 1. Project purpose

You will act as a cloud solutions team responding to a realistic service need. Choose a focused domain such as campus services, community wellbeing, retail operations, transport, sustainability, event management or small-business operations. Your solution must solve a clear user problem; technical features are valuable only when they support that problem.

The project integrates the module sequence: cloud fundamentals; IaaS/PaaS/SaaS and deployment choices; virtual machines, application deployment and scaling; cloud storage, databases and data analytics/AI; and security for cloud data access.

### 2. Learning outcomes assessed

| Outcome                               | What the team demonstrates                                                                                                |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Cloud architecture and service choice | Selects and justifies suitable cloud service and deployment models, including trade-offs.                                 |
| Cloud implementation and operations   | Deploys a working service using cloud resources, configuration, monitoring and an appropriate scaling/reliability design. |
| Cloud data and AI/ML                  | Uses cloud data storage/database services and implements a meaningful analytics or ML capability.                         |
| Cloud security                        | Applies least privilege, secure access, secret handling, network controls and data-protection measures.                   |
| Professional engineering practice     | Produces a reproducible, traceable submission with clear team contribution evidence.                                      |

### 3. Core challenge

Design and implement a secure, scalable cloud service that includes a user-facing web application or API and a data-driven capability. The service must use a cloud provider. AWS is recommended because it aligns with the laboratory activities; Azure or Google Cloud are acceptable if the required evidence can be supplied.

Your solution may start from an open-source baseline or AI-assisted code, provided you comply with the academic integrity policy, identify the source and licences, and explain the team's substantive modifications and decisions. You are assessed on engineering judgement, implementation and evidence—not on writing every line of code from scratch.

## 4. Minimum technical requirements

All requirements below are compulsory. Teams may choose implementation technologies that meet them.

| Area                       | Minimum requirement                                                                                                                                                                                                    | Acceptable evidence                                                                                                   |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Application                | A functioning web application or API with at least one meaningful user workflow and input validation.                                                                                                                  | Source code, local run instructions, API test collection or recorded test output.                                     |
| Compute and deployment     | Cloud deployment using a VM, managed application platform, container service or equivalent. Explain the chosen IaaS/PaaS/SaaS boundary.                                                                                | Infrastructure/configuration files, deployment record, architecture diagram and redacted provider evidence.           |
| Data layer                 | Persistent cloud storage or database with a defined schema/data model. No production credentials or personal data.                                                                                                     | Schema/migration, sample synthetic dataset, storage/database configuration and test output.                           |
| Analytics or AI/ML         | One data-driven feature that produces an interpretable output, e.g. dashboard metric, forecast, classifier, recommender, anomaly detection or retrieval/insight service.                                               | Notebook/script, dataset provenance, evaluation metric or validation results, example inputs/outputs and limitations. |
| Scalability and resilience | A documented approach to handling demand or component failure; implement at least one mechanism such as horizontal scaling, load balancing, queueing, caching, health checks, backup/recovery or managed availability. | Configuration, test procedure and measured/observed result.                                                           |
| Security                   | Authentication/authorisation appropriate to the service, least-privilege access, secure secret handling, restricted network/data access and validation against a named threat.                                         | Redacted policy/configuration, threat-control mapping, code/configuration and test evidence.                          |
| Operations                 | Logging/monitoring and one operational test or alert/query.                                                                                                                                                            | Dashboard/query export, logs with sensitive fields removed, and interpretation.                                       |

## 5. Required design and evaluation

### 5.1 Architecture and rationale

- Create one architecture diagram showing users, application components, cloud services, trust boundaries, data stores, observability and data flows. Use labels that match your repository and evidence filenames.
- Explain service-model and deployment-model choices, including at least two alternatives considered and why they were not selected.
- State key assumptions, constraints, expected workload and cost-control measures. Use synthetic or anonymised data only.

### 5.2 Test plan and results

Run and report at least four tests, including: (1) a functional workflow test, (2) a security control test, (3) a data/AI validation test, and (4) a scalability, resilience or recovery test. For each, record objective, setup, command/steps, expected result, actual result, date and artefact path. A failed test may still earn credit if it is honestly reported, diagnosed and followed by a sensible improvement plan.

### 5.3 Data and AI/ML expectations

The data/AI component must be more than a generic chatbot added to an unrelated application. It must use an identified dataset and help the stated user workflow or operational decision. Explain data provenance, fields/features, preprocessing, output interpretation, quality/evaluation approach, limitations and responsible-use considerations. Cloud-hosted inference is optional; a Colab or reproducible local notebook is acceptable when integrated with the service or its decision workflow.

## 6. Required submission package — designed for reliable AI-assisted marking

Submit one ZIP file named Group\_Gxxx\_INF2006\_Project.zip. Use the exact structure below. Do not submit passwords, access keys, tokens, personal data, private URLs or raw cloud credentials.

| Path                  | Purpose and required contents                                                                                                                                    |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| README.md             | Entry point: project title, one-paragraph problem statement, team members, quick-start commands, architecture image link, technology list and known limitations. |
| project_manifest.yaml | Machine-readable summary using the required schema in Appendix A. Every evidence item must have a stable relative path.                                          |
| report.pdf            | 8–12 pages excluding references and appendices. Use the required headings in Section 7. It must explain—not duplicate—the manifest.                              |
| src/                  | Runnable application, infrastructure/configuration and dependency files. Include a .env.example with placeholder values only.                                    |
| data/                 | Synthetic/sample data and a DATA_DICTIONARY.md. Do not include personal, sensitive or proprietary data.                                                          |
| analytics/            | Notebook(s) or script(s), requirements and a short README explaining how results are reproduced.                                                                 |
| evidence/             | Named, dated evidence artefacts cited by the manifest: test output, redacted screenshots, exported configuration, monitoring output and diagrams.                |
| tests/                | Automated tests, API collection or repeatable test scripts where feasible.                                                                                       |
| TEAM_CONTRIBUTIONS.md | Contribution matrix with one row per member: role, artefacts/commits, test/evidence ownership and reflection.                                                    |
| AI_USE_DECLARATION.md | Tools used, where they were used, sources/baselines, verification performed and licences/attribution.                                                            |
| video_link.txt        | Optional presentation link only. It is not primary assessment evidence.                                                                                          |

**Submission deadline:** Submit all project deliverables by **Sunday, 11 October, 11:59 PM (end of Week 6)**.

## 7. Required report headings

1. Problem, users and success criteria
2. Solution overview and architecture
3. Cloud service/deployment choices and trade-offs
4. Implementation, data design and security controls
5. Analytics or AI/ML feature: data, method, evaluation and limitations
6. Testing, scalability/resilience and monitoring results
7. Cost, sustainability and operational considerations
8. Team contribution, ethical considerations and reflection

## 8. project\_manifest.yaml — required schema

Use YAML exactly in this form. Teams may add fields but must not rename or remove the fields below. Paths must be relative to the ZIP root. Values must be factual and must match supplied artefacts.

```
project:
  group_id: "Gxxx"
  title: ""
  problem_statement: ""
  cloud_provider: "AWS | Azure | GCP | other"
  repository_commit: "optional commit hash"
team:
  members:
    - name: ""
      student_id: ""
      role: ""
architecture:
  diagram: "evidence/architecture.png"
  service_model: ""
  deployment_model: ""
  components: []
evidence:
  functional_test: "evidence/test-functional.md"
  security_test: "evidence/test-security.md"
  data_ai_test: "evidence/test-data-ai.md"
  scale_resilience_test: "evidence/test-resilience.md"
  monitoring: "evidence/monitoring.md"
data_ai:
  dataset: "data/README.md"
  method: ""
  evaluation: ""
  reproducible_command: ""
security:
  threat_control_map: "evidence/threat-control-map.md"
  secrets_handling: ""
run:
  prerequisites: []
  commands: []
declarations:
  ai_use: "AI_USE_DECLARATION.md"
  contributions: "TEAM_CONTRIBUTIONS.md"
```

## 9. Evidence rules

| Do                                                                                                                                                        | Do not                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Use a clear filename and refer to it from the manifest and report. Include commands, inputs and result summaries in text files.                           | Rely on a screenshot, video narration or live cloud URL as the only proof of a technical claim.                         |
| Redact account IDs, public IPs, URLs containing tokens and sensitive configuration values. Use screenshots only when text/config exports are unavailable. | Include credentials, production data, personal information, proprietary datasets or unlicensed code/assets.             |
| Make instructions runnable by a marker on a clean machine where feasible; state any unavoidable cloud prerequisites and expected cost.                    | Require the marker to log into a cloud account, join a team account, infer hidden configuration or make paid purchases. |

|                                                                        |                                                                                             |
|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| State limitations and failed tests honestly; link claims to artefacts. | Present AI-generated text/code as unverified fact or omit external code/AI-use attribution. |
|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|

## 10. Assessment rubric (100 marks)

| Criterion                                        | Marks | What earns strong marks                                                                                           | Evidence                                                                 |
|--------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Problem framing and architecture                 | 15    | Clear user need; coherent architecture; explicit service/deployment choices and trade-offs.                       | README, report sections 1–3, labelled architecture diagram and manifest. |
| Cloud implementation and functionality           | 20    | Working workflow; appropriate compute/network/data configuration; reproducible run/deploy instructions.           | src/, tests/, configuration and functional test record.                  |
| Cloud data and analytics/AI                      | 20    | Relevant data-driven capability; data provenance; reproducible method; credible evaluation and limitations.       | analytics/, data dictionary, output/evaluation and data/AI test record.  |
| Security and responsible cloud practice          | 15    | Threats mapped to controls; least privilege; secure secrets, access and validation; responsible data/AI practice. | Threat-control map, redacted configs, security test and declarations.    |
| Scalability, resilience and operations           | 15    | Implemented mechanism and meaningful test; monitoring evidence; sound interpretation and cost awareness.          | Scale/resilience test, monitoring output and report section 6–7.         |
| Engineering quality and reproducibility          | 10    | Well-organised code; documented dependencies; deterministic artefacts; evidence claims are traceable.             | ZIP structure, manifest validity, README, tests and evidence paths.      |
| Team contribution and professional communication | 5     | Specific, balanced contributions; concise report; clear attribution and reflection.                               | TEAM_CONTRIBUTIONS.md, AI_USE_DECLARATION.md and report.                 |

Marker principle: marks are awarded for verifiable evidence and sound reasoning. The marker will not infer functionality, security or contribution from ambition, unsupported claims, screenshots alone or a polished report. Missing or inaccessible evidence receives no credit for the associated claim.

To support consistent marking, use exact filenames, relative paths and the report headings stated above. A marker may run safe, offline commands described in your README. No submission will be penalised solely because cloud resources have been decommissioned, provided the required configuration and dated, redacted evidence are present.

## 11. Suggested 5–6 week plan

| Week | Milestone                                                                                                             |
|------|-----------------------------------------------------------------------------------------------------------------------|
| 1    | Confirm problem, users, success criteria, architecture and responsibilities. Create repository and manifest skeleton. |
| 2    | Implement core service, data model and secure access design.                                                          |
| 3    | Deploy cloud components; implement analytics/AI feature and reproducibility notes.                                    |
| 4    | Implement scaling/resilience/monitoring; execute tests and collect evidence.                                          |
| 5    | Close gaps, validate evidence paths, write report and conduct peer review.                                            |
| 6    | Final testing, packaging, submission preflight and presentation rehearsal.                                            |

## Appendix A. Submission preflight checklist

- ZIP filename follows the required convention.
- README, manifest, report, source, data/analytics, evidence and declarations are present.
- Every project\_manifest.yaml path exists and opens from the ZIP root.
- All required tests have objective, steps/command, expected result, actual result and artefact path.
- No credential, token, personal data or proprietary data remains in the submission.
- Architecture labels match component names in source/configuration and evidence.
- AI/baseline code, data sources and licences are declared; team verification is recorded.
- A teammate who did not create an artefact can follow the README and locate all evidence.